

6. (a) (i)

$$\Delta y = \frac{\lambda D}{a}$$

$$\frac{(4.0 - 0) \times 10^{-2}}{10} = \frac{\lambda(1.8)}{0.3 \times 10^{-3}}$$

$$\lambda = 6.666667 \times 10^{-7} \text{ m}$$

$$\approx 6.67 \times 10^{-7} \text{ m or } 667 \text{ nm}$$

1M+1M

1A

3

(ii) To ensure that the light through the 2 slits diffracts enough to interfere / overlap.

1A

1A

Explanation in terms of formula  
 $\lambda = \Delta y \frac{a}{D}$  is NOT accepted as  $a$  is the slit separation given, which is fixed.

2

(b) (i)

$$d \sin \theta = m\lambda$$

$$\frac{10^{-3}}{500} \sin \theta = 6.67 \times 10^{-7} \text{ m}$$

$$\theta = 19.471221^\circ \approx 19.47^\circ$$

Separation between central and first-order bright spot  
 $= 1.8 \tan 19.47^\circ$   
 $= 0.636396 \text{ m} \approx 0.636 \text{ m}$

1M

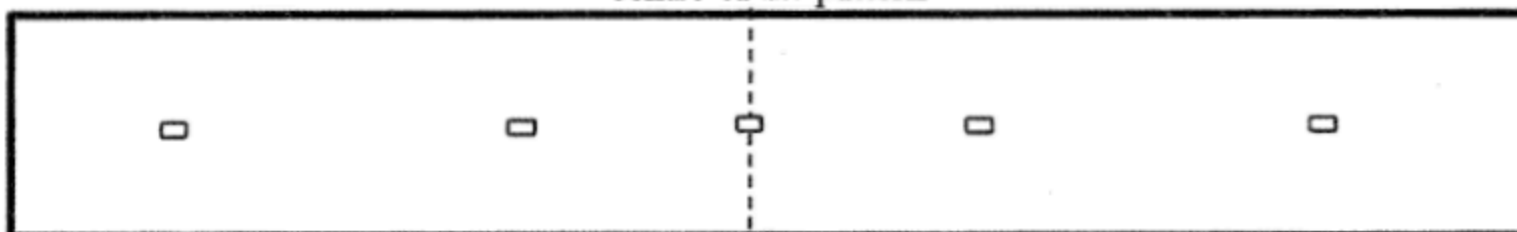
1M

1A

3

(ii)

centre of the pattern



symmetry about the central bright spot (2<sup>nd</sup> order shown)  
 separation between 2<sup>nd</sup> and 1<sup>st</sup> -order bright spots is larger

1A

1A

2